

MIAE 215 Final Exam Format – Winter 2025 Guide and Instructions Premium

Course: [Mechanical, Industrial & Aerospace engineering \(MIAE 215\)](#) 378 documents

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MIAE 215 Programming for Mechanical and Industrial Engineers Final Exam, April 30, 2025 (all sections)

- A two-sided formula sheet and a GCS approved calculator is permitted
- **Answer ALL questions on the question pages** in the space provided
- Write your name and ID number at the top of this page
- Do not separate the question pages or the **Arduino formula sheet** (last page)
- This exam has 12 pages

STUDENT NAME: _____

ID NUMBER: _____

Part I: Output Questions (40%)

Determine the outputs of the following programs and **write the answers in the answer boxes provided on the question pages**.

You can work outside the box or in the answer booklet. However, **only the answers inside the boxes will be marked**.

Answer any 4 of the 5 questions below. Each question is worth equal marks.

NOTE: If you answer all 5 questions we will mark only the first 4 questions.

Assume appropriate include and namespace std statements if needed. _____

Assume the programs have no compiler errors and they will produce output.

Q1)

// example output question

```
int main()
{
    cout << "\nhello world";

    return 0;
}
```

OUTPUT Q1:

1

Mock exam

Summary

Quiz

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Part II: Programming Questions (60%)

Answer the following questions **on the question pages** in the space provided.

Include and namespace statements are not required for the programs below. However, function prototypes are required in addition to function definitions.

The Arduino question **Q1 must be answered**. It is worth 10%.

Then **answer any TWO of the three questions from Q2 to Q4**. If more than two questions are answered then **only the first two questions will be marked**. Each question is worth 25%.

Q1) Write an Arduino program that ...

WRITE YOUR ANSWER FOR Q1) BELOW AND ON THE FOLLOWING PAGE

Q2) Write a C++ program that ...

Q3) Write a C++ program that ...

Q4) Write a C++ program that ...

WRITE YOUR ANSWERS FOR Q3) AND/OR Q4) ON THE FOLLOWING PAGES



void setup();
void loop();

Serial.begin(115200);
Serial.print("\ntext");

long unsigned int micros();
long unsigned int millis();
void delay(time_ms);

Servo servo1;
servo1.attach(pin);
servo1.write(angle_in_deg);

int analogRead(A0); // read analog pin A0 – typically returns 0 (0V) to 1023 (5V)

pinMode(q,OUTPUT); // configure DIO pin q to digital output
pinMode(q,INPUT_PULLUP); // configure DIO pin q to digital input

digitalRead(q); // read digital input pin q -- returns LOW (for 0V) and HIGH (for 5.0V)
digitalWrite(q,d); // set digital output pin q to d (LOW or HIGH)

ARDUINO FORMULA SHEET

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